

ECOLOGY 404B: ECOLOGY IN A DIGITAL AGE

Fall 2026 | Fully Face-to-Face | "Where Bayous Meet Bytes"

Instructor and Course Information

Item	Details
Instructor	Dr. Mary K. Beals, Ph.D.
Office	Fisher Hall 224A
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Email	mary.beals@sus.edu; response time: within 48 hours
Office Hours	By appointment and as announced in Canvas
Lecture	MWF 11:00-11:50 AM Fisher Hall 201
Lab	Friday 1:00-3:50 PM Fisher Hall 201
Course Delivery	Fully face-to-face lecture and lab
Course Theme	Where Bayous Meet Bytes

Course Description

Ecology 404B explores Louisiana ecological systems through field-connected inquiry, AI-supported learning, Science Gateway resources, and real-world problem solving. The course emphasizes populations, communities, ecosystems, and human-environment interactions while balancing scientific principles with creativity, storytelling, data interpretation, and applied problem-solving. In Fall 2026, the course is redesigned as a fully face-to-face lecture and lab experience with a course-based undergraduate research experience (CURE) centered on Louisiana ecosystems and environmental change.

Required Textbook and Materials

- **Textbook:** Molles Jr., M.C. & Simon, A.S. Ecology: Concepts & Applications, 9th ed. McGraw-Hill Connect eBook.
- **Learning Management System:** Canvas.
- **Course Collaboration Tools:** Padlet, NotebookLM, Notebook.ai, Microsoft Copilot, and instructor-approved AI tools.
- **Science Gateway/Computational Tools:** NAIRR CloudBank Classroom, Jupyter Notebooks/JupyterLab, Cornell Virtual Workshop, and HPC Carpentry tutorials.
- **Dataset Sources:** Instructor-curated, simulated, and public Louisiana-relevant ecological datasets, including wetland, biodiversity, climate, water quality, and coastal change data when appropriate.
- **Student Supplies:** Laptop or tablet for in-class/lab work when possible; notebook or digital field journal; reliable Canvas access; earbuds/headphones for tutorial work.

Course Goals and Learning Objectives

By the end of this course, students will be able to:

- Explain ecological concepts at the individual, population, community, and ecosystem levels.
- Apply ecological principles to real-world environmental challenges connected to Louisiana ecosystems.
- Use AI tools, simulations, CloudBank, Jupyter notebooks, and digital platforms for ecological inquiry.
- Develop reproducible workflow habits through file organization, documentation, data notes, and reflection.

- Analyze and visualize ecological data using guided, beginner-friendly computational workflows.
- Communicate scientific findings through data visualizations, written interpretation, presentation, and creative science communication.
- Demonstrate responsible AI and data use through transparency, human oversight, privacy protection, bias awareness, and accuracy checking.

Course Redesign Focus

This Fall 2026 redesign keeps the original "Where Bayous Meet Bytes" identity while shifting the course from a fully asynchronous format to a fully face-to-face lecture and lab structure. The redesign centers the lab around a CURE model so students practice ecological research in teams rather than only learning ecology as content. NAIRR CloudBank Classroom will serve as the primary student-facing workspace for project files, guided Jupyter notebook activities, and browser-based ecological data exploration. Cornell Virtual Workshop and HPC Carpentry resources will support student onboarding, workflow habits, and reproducibility.

CURE Structure: Digital Bayou Research Cycle

- Form a research team and complete a team charter.
- Identify a Louisiana-focused ecological issue or research question.
- Review background information and relevant ecological concepts.
- Explore instructor-curated, simulated, or public ecological datasets.
- Use CloudBank/Jupyter notebook activities to organize, visualize, and interpret data.
- Connect data patterns to ecological theory and environmental context.
- Communicate findings through the Digital Bayou Showcase.

Major Assignments

Assignment	Description	Category
Bayou Launch Team Charter	Teams define roles, communication norms, responsibilities, decision-making, and conflict-resolution plans.	Engagement/Lab
Digital Field Notes	Students document observations, ecological questions, dataset notes, notebook outputs, and reflections.	Creativity/Lab
CloudBank/Jupyter Onboarding	Students access CloudBank, organize files, open notebooks, and complete guided practice tasks.	Lab
Louisiana Dataset Exploration	Students inspect variables, metadata, data quality, and ecological meaning in a guided dataset.	Lab
CURE Research Question and Proposal	Teams develop a focused ecological question and explain the dataset/workflow they will use.	Lab/Engagement
Mid-Semester Data Checkpoint	Teams submit early visualizations, interpretation notes, and workflow documentation.	Lab
AI and Data Ethics Reflection	Students disclose tool use and evaluate AI/data limitations, bias, privacy, and accuracy.	Engagement/Creativity
Digital Bayou Showcase	Final team presentation with research question, data visualization, ecological interpretation, limitations, and next steps.	Creativity/Lab

Grading and Evaluation

Category	Weight	Examples
Exams	30%	Lecture exams, concept checks, data interpretation, and final project defense components.
Labs	30%	CURE labs, CloudBank/Jupyter notebooks, dataset exploration, data checkpoints, and lab deliverables.
Creativity	20%	Digital field notes, infographics, visual explanations, podcasts/media, and Digital Bayou Showcase products.
Engagement	20%	Attendance, preparation, team charter, in-class participation, peer review, Padlet/discussion activities, and reflections.

Grading Scale: A = 90-100 | B = 80-89 | C = 70-79 | D = 60-69 | F < 60

Face-to-Face Attendance and Participation

This course is fully face-to-face. Students are expected to attend lecture and lab in the scheduled delivery mode, arrive prepared, participate in team research work, and complete any missed work according to the instructor and university attendance policies. Lab sessions are essential because CloudBank/Jupyter onboarding, CURE workflow practice, research checkpoints, and team data interpretation will occur during scheduled lab time. Students who miss lab are responsible for communicating promptly, reviewing Canvas materials, and completing approved makeup work when applicable.

Late Work and Grace Tokens

Students may use two Grace Tokens for deadline extensions when permitted by assignment directions. After Grace Tokens are used, late work will not be accepted unless an approved university-related or documented excused absence applies. Grace Tokens cannot be used for exams, final presentations, or time-sensitive team obligations unless the instructor grants written approval.

Responsible AI Use and Disclosure

AI is welcome as a learning partner when it supports brainstorming, organization, clarification, grammar support, study planning, and feedback. AI may not replace student thinking, data interpretation, or required independent work. Students are responsible for all submitted work, including accuracy, originality, and ecological reasoning.

- Disclose the AI tool used, the prompt or input, and the output that influenced the submitted work.
- Use human oversight: review, revise, and verify AI-generated content before submitting.
- Protect privacy: do not submit personal, sensitive, or identifiable student data into public AI tools.
- Check for bias, missing context, hallucinated references, and ecological inaccuracies.
- Do not use AI to complete quizzes, exams, or restricted independent assessments unless explicitly authorized.

Data Ethics and Research Integrity

- Maintain accurate digital field notes and project records.
- Label files clearly and store project work in the assigned CloudBank/Jupyter workspace or Canvas-approved location.
- Do not fabricate, alter, or misrepresent data.

- Credit data sources, AI tools, tutorials, and team member contributions.
- Report unexpected problems, missing data, or analysis issues honestly.

Dr. Mary's Survival Guide

- Ask questions early. Confusion is easier to fix before the bayou gets muddy.
- Bring your laptop or device to lab when possible. If technology access is an issue, communicate early.
- Use Canvas as the official hub for announcements, assignment directions, files, and due dates.
- Keep your team charter alive. Revisit it when responsibilities, deadlines, or communication problems shift.
- Cite AI, cite data, cite people. Academic honesty is not optional.

Fall 2026 Course Schedule

Schedule follows the Fall 2026 Southern University calendar. Topics and due dates may be adjusted by Canvas announcement if needed.

Week	Dates	Theme & Topics	Key Activities / Assessments
1	Aug 17-21	Bayou Launch: Course Orientation, Ecology as Inquiry	CURE overview; Team Charter start; Digital field notebook setup
2	Aug 24-28	Mapping Our Roots: Natural History, Evolution, Louisiana Ecosystems	Local ecosystem observation; Louisiana ecological issue brainstorm
3	Aug 31-Sep 4	Researchable Questions and Ecological Evidence	CURE research question draft; literature/source matrix
4	Sep 7-11	Coding Resilience: Temperature, Water Stress, Climate Adaptation	No class Sep 7; CloudBank account check; Cornell Virtual Workshop onboarding
5	Sep 14-18	CloudBank/Jupyter Bootcamp	Jupyter notebook practice; file naming, markdown notes, basic visualization
6	Sep 21-25	Data in the Bayou: Variables, Metadata, and Data Quality	Louisiana dataset exploration; metadata and data quality checklist
7	Sep 28-Oct 2	Data-Driven Populations: Population Growth and Life Histories	Population notebook; sampling or species occurrence analysis
8	Oct 5-9	Mid-Semester Synthesis and Exam Week	Midterm exam/data checkpoint; team progress conference
9	Oct 12-16	Networks and Niches: Competition, Mutualism, Community Ecology	Fall Break Oct 15-16; no Friday lab; CURE proposal revision
10	Oct 19-23	Community Patterns and Species Interactions	Invasive species, pollinator, or interaction network activity
11	Oct 26-30	Systems in Motion: Nutrient Cycles and Succession	Nutrient/succession data story; HPC Carpentry workflow habit check
12	Nov 2-6	Visualization and Ecological Interpretation	Clean figures; captions; interpretation paragraphs; peer feedback
13	Nov 9-13	Climate in the Cloud: Climate Change and Landscape Ecology	Coastal change/resilience notebook; Baton Rouge 2050 connection
14	Nov 16-20	From Data to Claim: Results, Limitations, and Meaning	Draft results/discussion; AI and data ethics reflection
15	Nov 23-27	Thanksgiving Week: Team Data Checkpoint	No lab Nov 27; asynchronous team update or Canvas checkpoint as assigned
16	Nov 30-Dec 4	Digital Bayou Showcase	Final presentations; project defense; final field notebook due
Finals	Dec 7-11	Final Examination Period	Final exam/project defense window as scheduled by university/instructor

Institutional Policies

Students are expected to follow Southern University and A&M College policies related to student conduct, academic integrity, academic grievances, ADA/disability services, Title IX/non-discrimination, student email, course evaluations, and attendance. The instructor reserves the right to make changes to the syllabus, course content, or instructional techniques when necessary. Written notice of changes will be provided through SUS email and/or Canvas announcements.

Selected References and Course Resource Links

- Molles Jr., M.C. & Simon, A.S. Ecology: Concepts & Applications, 9th ed. McGraw-Hill Connect eBook.
- Southern University and A&M College Fall 2026 Academic Calendar: <https://www.subr.edu/page/academic-affairs-academic-calendar>
- NAIRR CloudBank Classroom: <https://www.cloudbank.org/training/nairr-pilot-cloudbank-classroom>
- Project Jupyter Documentation: <https://docs.jupyter.org/>
- Cornell Virtual Workshop: <https://cvw.cac.cornell.edu/>
- HPC Carpentry Lessons: <https://www.hpc-carpentry.org/lessons/>
- NOAA Digital Coast: <https://coast.noaa.gov/digitalcoast/>
- USGS Water Data for the Nation: <https://waterdata.usgs.gov/>
- GBIF: <https://www.gbif.org/>